ICE (Ice Cream Experiments) lesson

by Norman Brooks

For the first session, students will be given ingredients and instructions for making ice cream. In subsequent sessions they will bring other ingredients to create new flavors, improve on recipes, and improve the quality of their ice cream. As the quality improves, they will want to be able to replicate or produce it consistently. This will require accurate measurements.

OBJECTIVES:

The Beauty of this lesson is that it can be used to reach so many objectives. It is most often used as an activity to teach, give practice, or to assess students' knowledge of dealing with variables. It also offers exposure to the experimental process, forms of matter, changing forms of matter and many other science concepts and processes.

INGREDIENTS and MATERIALS:

(For a class of 30 students who work in pairs.)

- 2-4 bags of party ice
- 1 gallon whole milk
- ice chest
- 1 box rock salt
- 8 oz paper cups (36)
- 5 lbs. sugar
- 1 gallon zip lock bags (15)
- vanilla extract
- plastic spoons (36)
- 1 quart zip lock bags (15)
- paper towels
- two trash cans
- Medicine droppers for extract
- plastic scoops for ice/salt

Trash cans are for disposal of bags, spoons, and cups. Teachers can decide which to recycle. Doing this activity outside simplifies clean up.

PROCEDURE:

1. Combine Sugar (2-4 tblspoons), Extract (a few drops), Milk 8 oz. per pair) , in small zip lock bag and zip it up. This is the **S E M bag**
2. Combine ice (2 cups) and rock salt (1/2 cup), in large zip lock bag.**This is the S I bag**
3. Insert **S E M bag** inside the **S I bag** and zip it up.
4. Each partner takes one end of the S I bag and the two shake until ice cream is firm.
5. Cups and spoons are then distributed to students for taste testing and eating.

VARIABLES

All ingredients and supplies are VARIABLES and can be varied. The students should be encouraged to experiment by changing variables for different results. To conduct an effective experiment, they should only change one variable at a time, and should keep accurate records of what they have done, and what the results were.

SOME EXAMPLES OF LESSON EXTENSIONS:

- Rubrics for judging results
- History of ice cream
- Different flavors
- Fractions and proportions
- Improving appearance

- Ice cream in different cultures
- What's in commercial ice cream?
- How much fat is in ice creams?
- Changing the consistency
- Economics (costs)
- Following directions
- Working in pairs
- Changing forms of matter
- Why salt?